

SQL Window Functions

Design following tables and solve questions using window functions

- demand2 → day, qty
- demand → product, day, qty
- Sales data → product, location, sales, refunds

Demand2 table

day	qty
1	10
2	6
3	21
4	9
6	12
7	18
8	3
9	6
10	23

Demand table

product	day	qty
A	1	10
A	2	6
A	3	21
A	4	9
A	5	19
B	1	12
B	2	18
B	3	3
B	4	6
B	5	23

Sales data

product	day	qty
A	1	10

product	day	qty
A	2	6
5	19	21
1	12	9
2	18	
B		
B		
B	3	3
B	4	6
B	5	23

SQL Window Functions – Questions

- Write a query to display the `day` and `qty` from the `demand2` table along with the **previous day's quantity** using a window function.
- Write a query to display the `day`, `qty`, and the **next day's quantity** using a window function.
- Write a query to display the `day`, `qty`, and the **difference between the current day's quantity and the previous day's quantity**.
- Write a query to display the `product`, `day`, `qty`, and the **previous day's quantity for each product** using a window function.
- Write a query to display the `product`, `day`, `qty`, and the **next day's quantity for each product**.
- Write a query to display the `product`, `day`, `qty`, and the **difference between the current day's quantity and the previous day's quantity for each product**.
- Write a query to display the `day`, `qty`, and the **cumulative sum of quantity** using a window function.
- Using the `sales_data` table, write a query to display `product`, `location`, `sales`, `refunds`, and **net sales (`sales` - `refunds`)**.
- Write a query to **rank the products based on sales within each location** using a window function.
- Write a query to **calculate the total sales from the top 3 best-selling products in each location**.
- Write a query to display each product's sales along with the **total sales of its location** using a window function.
- Write a query to display each product's sales along with the **percentage contribution of that product to the total sales of its location**.
- Write a query to display the **running total of sales for each location** using a window function.

14. Write a query to display each product's sales along with the **previous product's sales within the same location**.
15. Write a query to display each product's sales along with the **difference between its sales and the previous product's sales within the same location**.